Exercise 5

AIM :

To write a C program to perform below actions in an array element with length of 10.

1. How many even Numbers in the array
2. sort the array in ascending order
3. sort the array in descending order
4. remove repetition numbers and print the array

ALGORITHM :

STEP 1: INITIALIZATION

1. DECLARE variables:

   - arr[10] : Original array
   - ascending[10] : For ascending sorted array
   - descending[10]: For descending sorted array
   - i : Loop counter

2. DISPLAY "Enter 10 integer elements:"

3. FOR i = 0 to 9:

   3.1 READ integer input

   3.2 STORE in arr[i]

STEP 2: DISPLAY ORIGINAL ARRAY

4. DISPLAY header: "=========ARRAY OPERATIONS========="

5. DISPLAY "Original Array: "

6. CALL displayArray(arr, 10)

   - FOR each element in arr:

     PRINT element followed by space

   - PRINT newline

STEP 3: COUNT EVEN NUMBERS

7. DISPLAY "1. Even Numbers Count: "

8. CALL countEvenNumbers(arr, 10):

   8.1 INITIALIZE count = 0

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  8.2 FOR i = 0 to 9:
      IF arr[i] % 2 == 0:
          count = count + 1
  8.3 RETURN count
9. DISPLAY returned count
```

STEP 4: ASCENDING SORT

```
10. COPY original array to ascending array:
    FOR i = 0 to 9:
        ascending[i] = arr[i]

11. DISPLAY "2. Ascending Order: "
12. CALL sortAscending(ascending, 10):
    12.1 FOR i = 0 to 8:
        12.1.1 FOR j = 0 to 8-i:
               IF ascending[j] > ascending[j+1]:
                  - temp = ascending[j]
                  - ascending[j] = ascending[j+1]
                  - ascending[j+1] = temp
13. CALL displayArray(ascending, 10) to show sorted array
```

STEP 5: DESCENDING SORT

```
14. COPY original array to descending array:
    FOR i = 0 to 9:
        descending[i] = arr[i]

15. DISPLAY "3. Descending Order: "
16. CALL sortDescending(descending, 10):
    16.1 FOR i = 0 to 8:
        16.1.1 FOR j = 0 to 8-i:
               IF descending[j] < descending[j+1]:
                  - temp = descending[j]
                  - descending[j] = descending[j+1]
                  - descending[j+1] = temp
17. CALL displayArray(descending, 10) to show sorted array
```

STEP 6: REMOVE DUPLICATES

```
18. DISPLAY "4. Without Duplicates: "
19. CALL removeDuplicates(arr, 10):
    19.1 FOR i = 0 to 9:
        19.1.1 SET isDuplicate = 0 (false)
        19.1.2 FOR j = 0 to i-1:
               IF arr[i] == arr[j]:
                  - SET isDuplicate = 1 (true)
                  - BREAK inner loop
        19.1.3 IF isDuplicate == 0:
```

PRINT arr[i] followed by space

19.2 PRINT newline

STEP 7: PROGRAM TERMINATION

20. RETURN 0 (successful execution)

21. END

OUTPUT :

```
D:\C programming\DHAVAMANI_EX5>gcc main.c -o arr.exe

D:\C programming\DHAVAMANI_EX5>arr.exe
Enter 10 integer elements:
97 98 94 46 97 34 45 56 96 34

=========ARRAY OPERATIONS=========

Original Array: 97 98 94 46 97 34 45 56 96 34

1. Even Numbers Count: 7
2. Ascending Order: 34 34 45 46 56 94 96 97 97 98
3. Descending Order: 98 97 97 96 94 56 46 45 34 34
4. Without Duplicates: 97 98 94 46 34 45 56 96
```